ECON2103 Microeconomics

Chapter 7 The Cost of Production

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Outline

1. Measuring Cost: Which Costs Matter?
2. Cost in the Short Run
3. Cost in the Long Run
4. Long-Run versus Short-Run Cost
5. Production with Two Outputs - Economies of Scope
6. Summary and Exercises

Economic Cost versus Accounting Cost

- **accounting cost** - Actual expenses plus depreciation charges for capital equipment.
- **economic cost** - Cost to a firm of utilizing economic resources in production.
- **opportunity cost** - Cost associated with opportunities forgone when a firm's resources are not put to their best alternative use.

$$\text{Economic cost} = \text{Opportunity cost}$$

Sunk Cost

sunk cost - Expenditure that has been made and cannot be recovered.

- Because a sunk cost cannot be recovered, it should not influence the firm's decisions.
- If specialized equipment for a plant cannot be converted for alternative use, the expenditure on this equipment is a sunk cost, and should not be included as part of the firm's economic costs.
- A *prospective* sunk cost is an *investment*.

Fixed Costs and Variable Costs

- **total cost (TC OR C)** - Total economic cost of production, consisting of fixed and variable costs.
- **fixed cost (FC)** - Cost that does not vary with the level of output and that can be eliminated only by shutting down.
- **variable cost (VC)** - Cost that varies as output varies.
- Fixed cost does not vary with the level of output. The only way that a firm can eliminate its fixed costs is by shutting down.
    - Shutting down doesn't necessarily mean going out of business.

Fixed or Variable?

- Over a very short time horizon - say, a few months - most costs are fixed.
- Over a very long time horizon - say, ten years - nearly all costs are variable.

Fixed versus Sunk Costs

- Fixed costs can be avoided if the firm shuts down a plant or goes out of business. Sunk costs, on the other hand, are costs that have been incurred and cannot be recovered.
- Fixed costs affect the firm's decisions looking forward, whereas sunk costs do not.
- Incurring a high sunk cost might later turn out to be a bad decision, but the expenditure is gone and cannot be recovered by shutting down.
- A *prospective* sunk cost is different and does affect the firm's decisions looking forward.

Amortizing Sunk Costs

amortization - Policy of treating a one-time expenditure as an annual cost spread out over some number of years.

Amortizing large capital expenditures and treating them as ongoing fixed costs can simplify the economic analysis of a firm's operation.

For simplicity, we will usually treat sunk costs in this way as we examine the firm's production decision.

Example: Sunk, Fixed, and Variable Costs

- **Computers**: Most costs are variable.
  - Most important are the cost of components and labor.
- **Software**: Most costs are sunk.
  - A software firm will spend a large amount of money to develop a new application.
- **Pizzas**: Most costs are fixed.
  - Such as the opportunity cost of the owner's time, rent, and utilities.

Marginal and Average Cost

TABLE 7.1 A FIRM'S COSTS

RATE OF OUTPUT (UNITS PER YEAR)	FIXED COST (DOLLARS PER YEAR)	VARIABLE COST (DOLLARS PER YEAR)	TOTAL COST (DOLLARS PER YEAR)	MARGINAL COST (DOLLARS PER UNIT)	AVERAGE FIXED COST (DOLLARS PER UNIT)	AVERAGE VARIABLE COST (DOLLARS PER UNIT)	AVERAGE TOTAL COST (DOLLARS PER UNIT)
	(FC) (1)	(VC) (2)	(TC) (3)	(MC) (4)	(AFC) (5)	(AVC) (6)	(ATC) (7)
0	50	0	50	—	—	—	—
1	50	50	100	50	50	50	100
2	50	78	128	28	25	39	64
3	50	98	148	20	16.7	32.7	49.3
4	50	112	162	14	12.5	28	40.5
5	50	130	180	18	10	26	36
6	50	150	200	20	8.3	25	33.3
7	50	175	225	25	7.1	25	32.1
8	50	204	254	29	6.3	25.5	31.8
9	50	242	292	38	5.6	26.9	32.4
10	50	300	350	58	5	30	35
11	50	385	435	85	4.5	35	39.5

Marginal and Average Cost (Con't)

- **marginal cost (MC)** - Increase in cost resulting from the production of one extra unit of output.

$$MC = \Delta VC/\Delta q = \Delta TC/\Delta q$$

- **average cost**
  - **average total cost (ATC)** - Firm's total cost divided by its level of output

$$ATC = TC/q$$

  - **average fixed cost (AFC)** - Fixed cost divided by the level of output.

$$AFC = FC/q$$

  - **average variable cost (AVC)** - Variable cost divided by the level of output.

$$AVC = VC/q$$

The Determinants of Short-Run Cost

Because $\Delta VC = w\Delta L$, it follows that

边际成本（MC） $MC = \Delta VC/\Delta q = w\Delta L/\Delta q$

where w is the fixed wage and ΔL is the extra output.

Recall that $MP_L = \Delta q/\Delta L$. As a result,

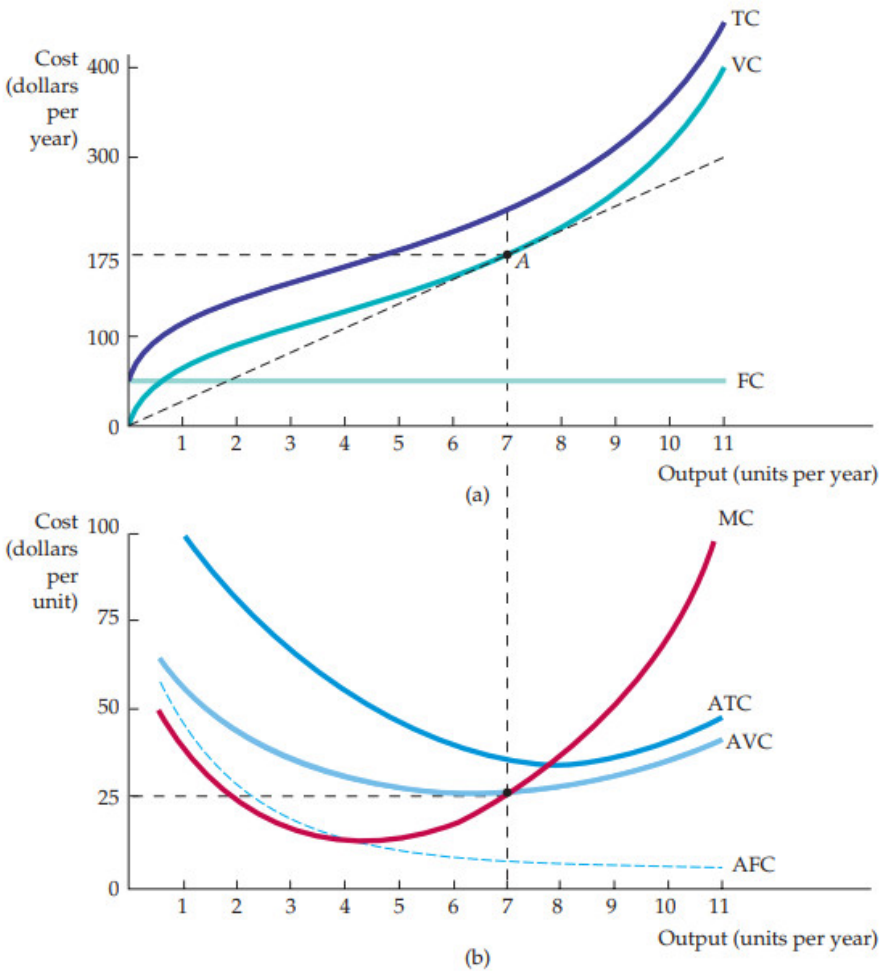
劳动边际产量（MPL） $MC = w/MP_L$

Diminishing Marginal Returns and Marginal Cost

- Diminishing marginal returns means that the marginal product of labor declines as the quantity of labor employed increases.
- When there are diminishing marginal returns, marginal cost will increase as output increases.

The Shapes of the Cost Curves

这表明边际成本等于总成本曲线在某一点的斜率



- In (a) TC = FC + VC.
- In (b) ATC = AVC + AFC.
- MC crosses AVC and ATC at their minimum points.

The Average-Marginal Relationship

- The vertical distance between the ATC and AVC curves decreases as output increases.
- The minimum point of the ATC curve must lie above and to the right of the minimum point of the AVC curve.
- The tangent to the VC curve at A is the marginal cost of production. At A, MC = AVC because AVC is minimized at this output.

Example: The Short-Run Cost of Aluminum Smelting

The expenditure on a smelting plant, although substantial, is a sunk cost and can be ignored. Fixed costs are relatively small and can also be ignored.

TABLE 7.2 PRODUCTION COSTS FOR ALUMINUM SMELTING ($/TON) (BASED ON AN OUTPUT OF 600 TONS/DAY)

PER-TON COSTS THAT ARE CONSTANT FOR ALL OUTPUT LEVELS	OUTPUT ≤ 600 TONS/DAY	OUTPUT > 600 TONS/DAY
Electricity	$316	$316
Alumina	369	369
Other raw materials	125	125
Plant power and fuel	10	10
Subtotal	$820	$820
PER-TON COSTS THAT INCREASE WHEN OUTPUT EXCEEDS 600 TONS/DAY		
Labor	$150	$225
Maintenance	120	180
Freight	50	75
Subtotal	$320	$480
Total per-ton production costs	**$1140**	**$1300**

Example: The Short-Run Cost of Aluminum Smelting (Con't)

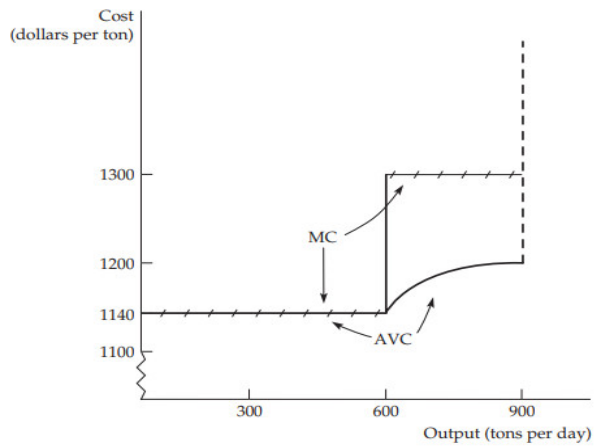
When $q < 600$, $\text{TVC} = 1140q$, so MC = AVC = 1140.

When $q > 600$, total variable cost is given by:

$$TVC = (1140)(600) + 1300(q - 600) = 1300q - 96000$$

Therefore average variable cost is

$$AVC = 1300 - \frac{96000}{q}$$



The User Cost of Capital

user cost of capital - Annual cost of owning and using a capital asset, equal to economic depreciation plus forgone interest.
经济折旧 机会成本

User Cost of Capital = Economic Depreciation + (Interest Rate)(Value of Capital)

We can also express the user cost of capital as a *rate* per dollar of capital:

$$r = \text{Depreciation rate} + \text{Interest rate}$$
经济折旧率 利息率或资本的机会成本

The Cost-Minimizing Input Choice

- **The price of capital**
  - We amortize the expenditure by spreading it over the lifetime of the capital and account for the forgone interest.
  - The price of capital is its user cost: r = Depreciation rate + Interest rate.
- **The rental rate of capital**
  - **rental rate** - Cost per year of renting one unit of capital.
  - If the capital market is competitive, the rental rate should be equal to the user cost, r.
  - Capital that is purchased can be treated as though it were rented as a rental rate equal to the user cost of capital.

The Isocost Line

isocost line - Graph showing all possible combinations of labor and capital that can be purchased for a given total cost.

The Total cost C of producing any particular output is given by the sume of the firm's labor cost and its capital cost:

$$C = wL + rK$$

C: 总成本;
w: 单位劳动的工资率
L: 劳动投入量;
r: 资本的租赁率或使用成本;
K: 资本投入量。

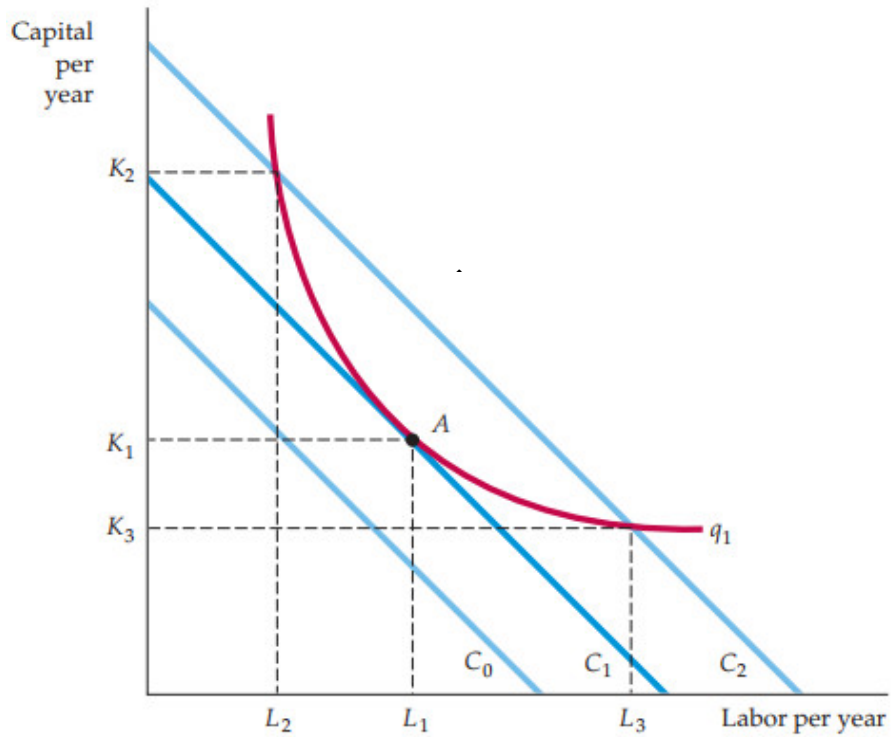
If we rewrite the total cost equation as an equation for a straight line, we get

$$K = C/r - (w/r)L$$

It follows that the isocost line has a slope of $\Delta K/\Delta L = -(w/r)$, which is the ratio of the wage rate to the rental cost of capital.

Choosing Inputs

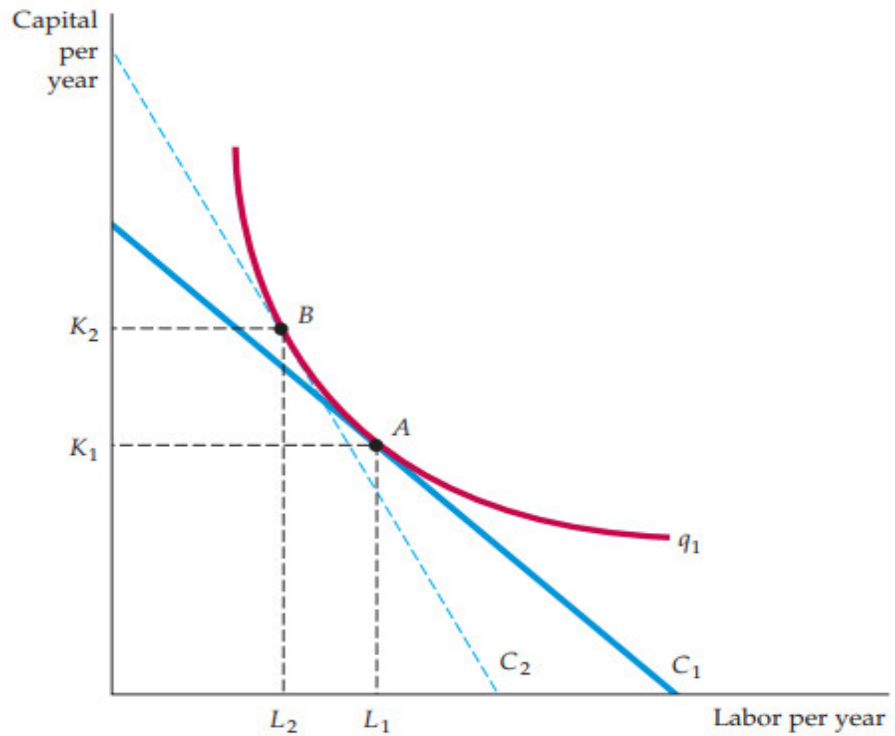
Producing a Given Output at Minimum Cost



Choosing Inputs (Con't)

Input Substitution When an Input Price Changes

When the price of labor increases, the isocost curves become steeper.



Choosing Inputs (Con't)

Since

$$\text{MRTS} = -\Delta K/\Delta L = MP_L/MP_K$$

It follows that when a firm minimizes the cost of producing a particular output, the following condition holds:

$$MP_L/MP_K = w/r$$

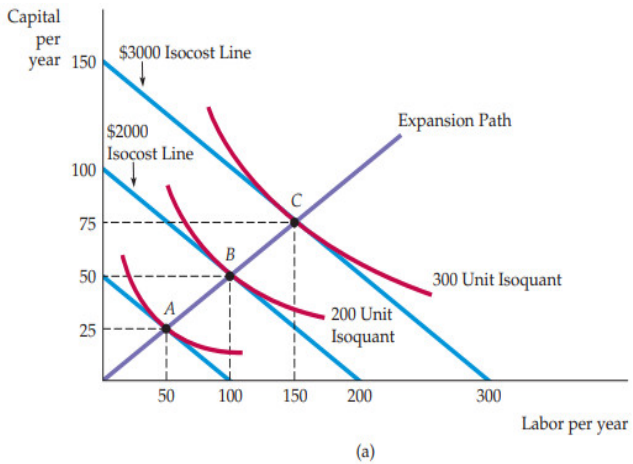
\Delta K：资本的减少量;
\Delta L：劳动的增加量;
MPL：劳动的边际产量;
MPK：资本的边际产量。

We can rewrite this condition as follows:

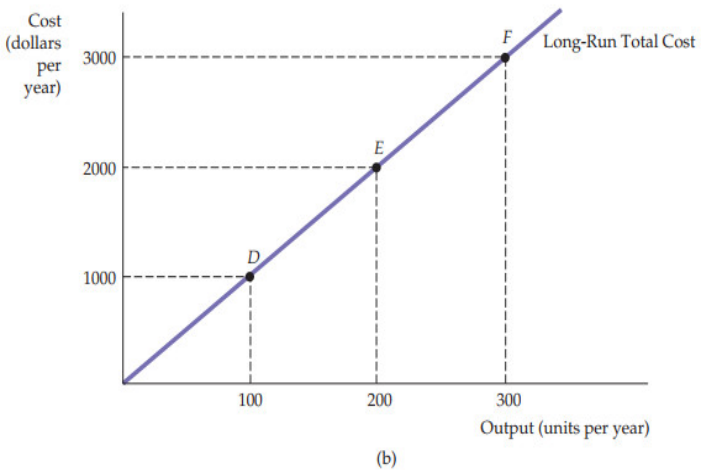
$$MP_L/w = MP_K/r$$

Cost Minimization with Varying Output Levels

expansion path - Curve passing through points of tangency between a firm's isocost lines and its isoquants.



扩展路径是连接企业等成本线（Isocost Lines）与等产量曲线（Isoquants）切点的曲线。表示企业在不同产量水平下实现成本最小化的劳动和资本组合



扩展路径与 LRTC 的关系：
扩展路径的每个点提供了长期总成本曲线的一个数据点。
扩展路径包含了 LRTC 曲线的所有信息。

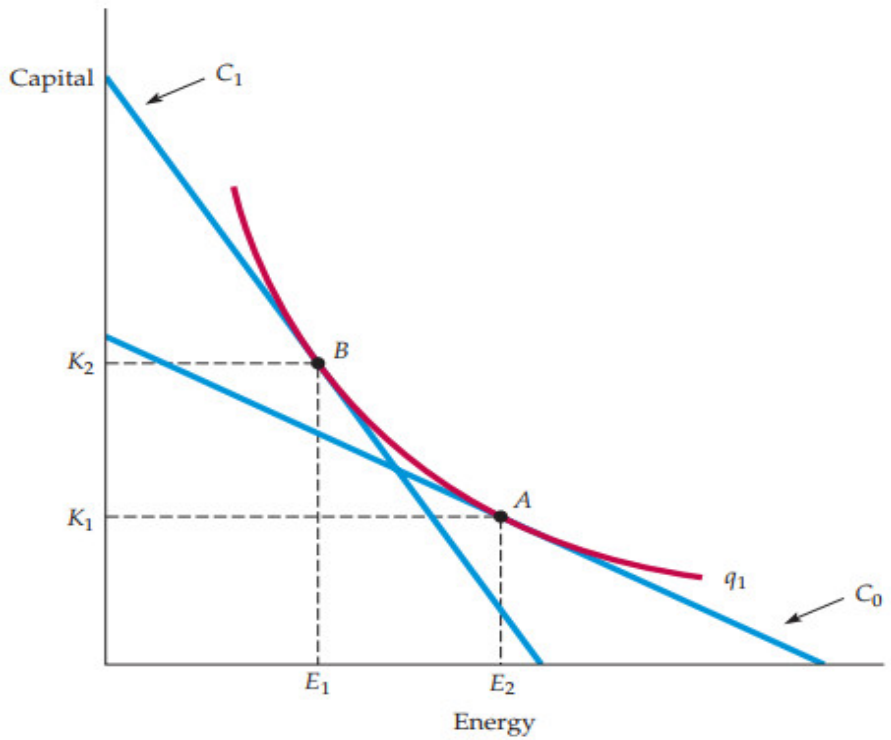
Expansion Path and Long-Run Costs

The firm's expansion path contains the same information as its long-run total cost curve.

- The expansion path illustrates the lowest-cost combinations of labor and capital that can be used to produce each level of output in the long run - i.e., when both inputs to production can be varied.
- The long-run total cost curve measured the least cost of producing each level of output.

Example: Reducing the Use of Energy

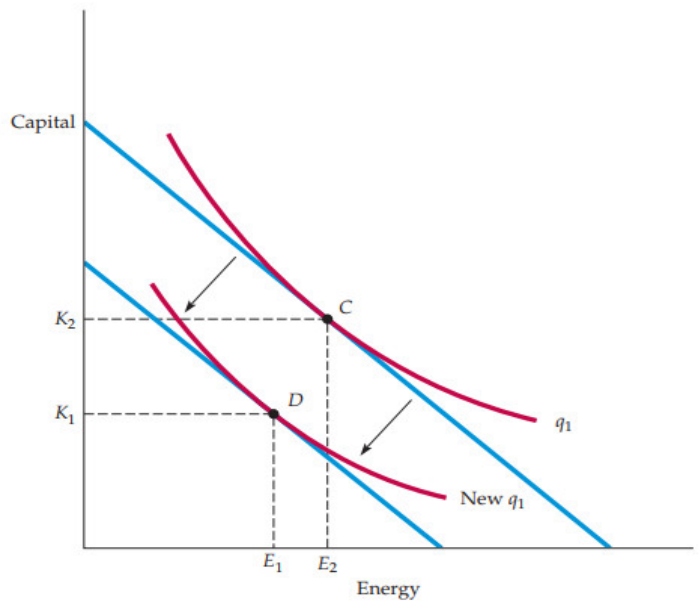
Greater energy efficiency can be achieved if capital is substituted for energy.



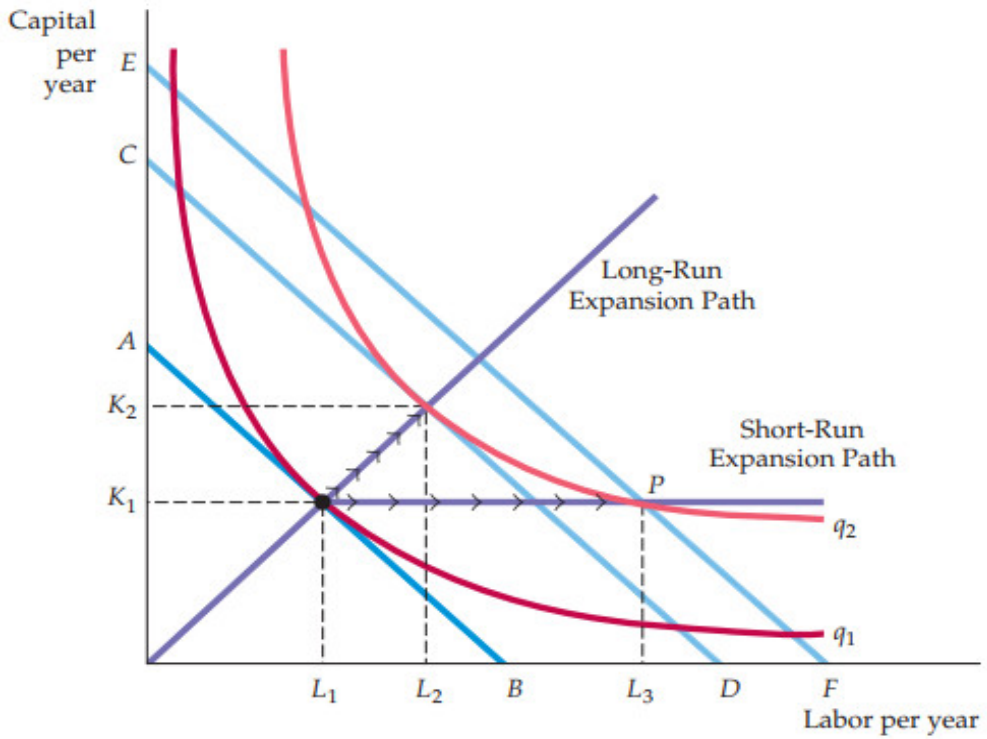
Example: Reducing the Use of Energy (Con't)

Greater energy efficiency can be achieved through technological change.

- The same output can be produced with smaller amounts of inputs.
- The isoquant shifts inward. The same output can be produced with less energy and capital.



The Inflexibility of Short-Run Production



When a firm operates in the short run, its cost of production may not be minimized because of inflexibility in the use of capital inputs.

Long-Run Average Cost

- **long-run average cost curve (LAC)** - Curve relating average cost of production to output when all inputs, including capital, are variable.
- **short-run average cost curve (SAC)** - Curve relating average cost of production to output when level of capital is fixed.
- **long-run marginal cost curve (LMC)** - Curve showing the change in long-run total cost as output is increased incrementally by 1 unit.

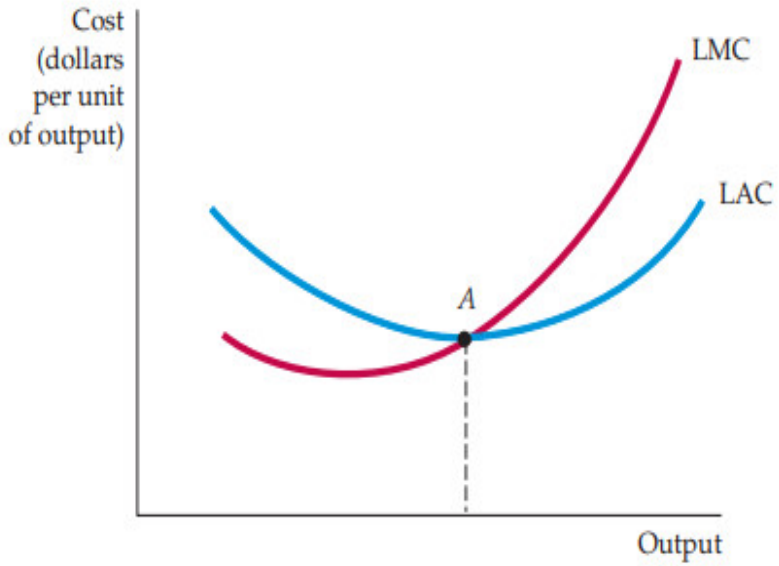
Long-Run Average and Marginal Cost

- When a firm is producing at an output at which the long-run average cost LAC is falling, the long-run marginal cost LMC is less than LAC.
- When LAC is increasing, LMC is greater than LAC.
- The two curves intersect at *A*, where the LAC curve achieves its minimum.

当 LAC 下降时（规模经济阶段），LMC < LAC：
表明每增加一单位产出的额外成本（LMC）低于平均成本（LAC），从而拉低了 LAC。

当 LAC 上升时（规模不经济阶段），LMC > LAC：
表明额外产出的成本高于平均成本，推动 LAC 上升

LMC 与 LAC 的交点：
LMC 和 LAC 在 A 点相交，该点是 LAC 的最低点。
此时产出规模达到最优规模（Optimal Scale），长期平均成本最低。



Economies and Diseconomies of Scale

As output increases, the firm's average cost of producing that output is likely to decline, at least to a point. This can happen for the following three reasons:

1. If the firm operates on a larger scale, workers can specialize in the activities at which they are most productive.
2. Scale can provide flexibility. By varying the combination of inputs utilized to produce the firm's output, managers can organize the production process more effectively.
3. The firm may be able to acquire some production inputs at lower cost because it is buying them in large quantities and can therefore negotiate better prices.

Economies and Diseconomies of Scale (Con't)

At some point, however, it is likely that the average cost of production will begin to increase with output. There are three reasons for this shift:

1. At least in the short run, factory space and machinery may make it more difficult for workers to do their jobs effectively.
2. Managing a larger firm may become more complex and inefficient as the number of tasks increases.
3. The advantages of buying in bulk may have disappeared once certain quantities are reached. At some point, available supplies of key inputs may be limited, pushing their costs up.

Economies and Diseconomies of Scale (Con't)

- **economies of scale** - Situation in which output can be doubled for less than a doubling of cost.
- **diseconomies of scale** - Situation in which a doubling of output requires more than a doubling of cost.
- The term *economies of scale* includes increasing returns to scale as a special case, but it is more general because it reflects input proportions that change as the firm changes its level of production

Increasing Returns to Scale: Output more than doubles when the quantities of all input are doubled.

指当所有投入翻倍时，产量增加超过两倍。
是技术层面的概念，更多关注生产函数本身的特性。

Economies of Scale: A doubling of output requires less than a doubling of cost.

指在产量增加的同时，平均成本下降。
是经济层面的概念，涉及到成本和价格等市场因素。

Measurement of Economies of Scale

Economies of scale are often measured in terms of a cost-output elasticity, E_C.

成本-产出弹性

$$E_C = (\Delta C/C)/(\Delta q/q)$$

Rewrite the equation as follows:

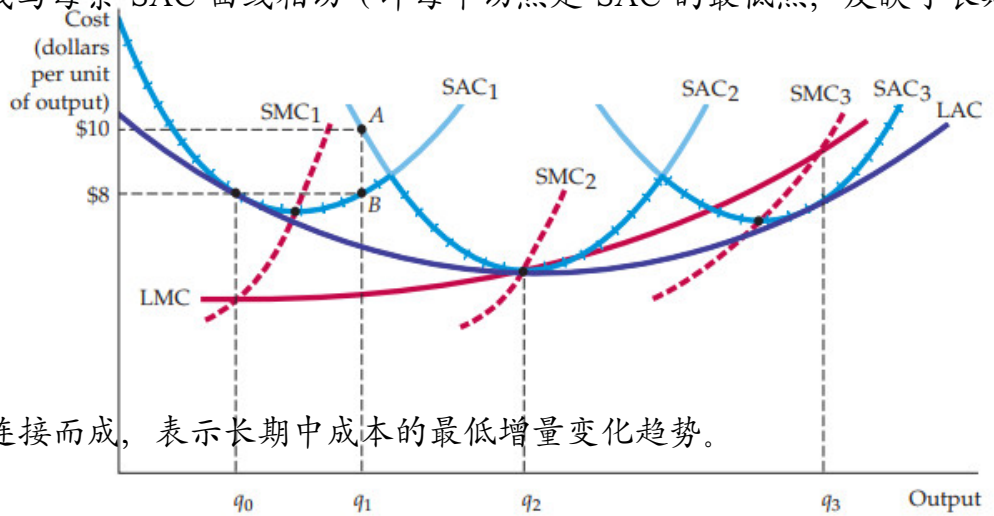
$$E_C = (\Delta C/\Delta q)/(C/q) = MC/AC$$

边际成本是“新增成本”，而平均成本是“历史平均”。
当新增的边际成本 MC 比当前的平均成本 AC 低时，新增加的这一部分成本会拉低总成本的平均值，导致 AC 下降。

- When there are economies of scale, $E_C < 1$.
- When there are diseconomies of scale, $E_C > 1$.
- When $E_C = 1$, there are neither economies nor diseconomies of scale.

The Relationship between Short-Run and Long-Run Cost

包络线（Envelope Curve）是一条由一组曲线的外边界形成的曲线
LAC 是所有 SAC 中最低点的集合
LAC 曲线与每条 SAC 曲线相切（即每个切点是 SAC 的最低点，反映了长期最优的资本与劳动组合）。



LMC 通过每条 SMC 的最低点连接而成，表示长期中成本的最低增量变化趋势。

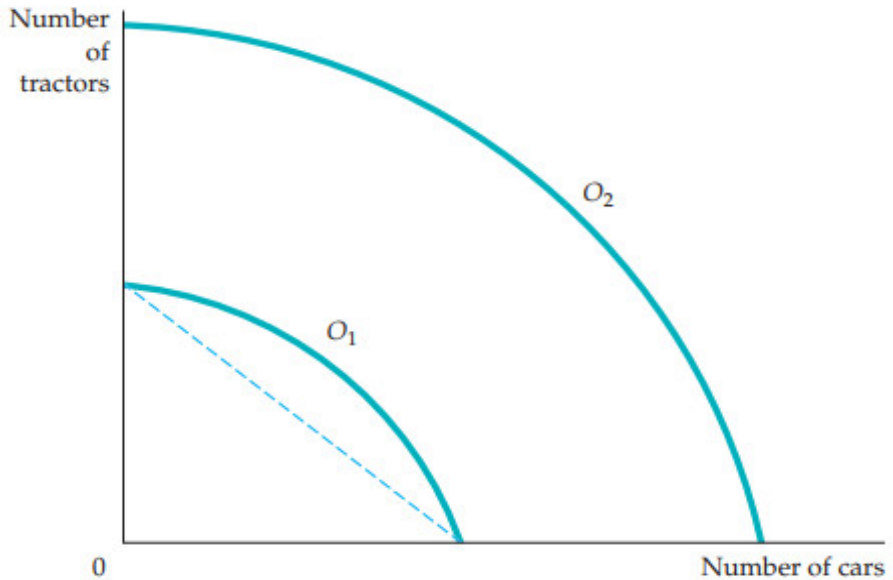
每一种固定资本水平对应一条短期平均成本曲线（SAC）。
短期成本受固定资本约束，可能无法实现成本最小化。

4.规模经济与不经济:
LAC 的下降阶段（LMC < LAC）对应规模经济;
LAC 的上升阶段（LMC > LAC）对应规模不经济。

- The long-run average cost curve *LAC* is the envelope of the short-run average cost curves SAC_1, SAC_2, and SAC_3.
- With economies and diseconomies of scale, the minimum points of the short-run average cost curves do not lie on the long-run average cost curve.
- Each point on the long-run marginal cost curve is the short-run marginal cost associated with the most cost-efficient plant.

Product Transformation Curve

product transformation curve - Curve showing the various combinations of two different outputs (products) that can be produced with a given set of inputs.



当企业同时生产两种产品时，比单独生产其中一种产品更节省资源。
资源可以在两种产品之间高效分配，减少生产成本

The product transformation curves O_1 and O_2 are bowed out (or concave) because there are economies of scope in production.

Economies and Diseconomies of Scope

- **economies of scope** - Situation in which joint output of a single firm is greater than output that could be achieved by two different firms when each produces a single product.
- **diseconomies of scope** - Situation in which joint output of a single firm is less than could be achieved by separate firms when each produces a single product.
- **degree of economies of scope (SC)** - Percentage of cost savings resulting when two or more products are produced jointly rather than Individually.

$$SC = \frac{C(q_1) + C(q_2) - C(q_1, q_2)}{C(q_1, q_2)}$$

当 $SC > 0$，表示存在范围经济;
当 $SC = 0$，表示既没有范围经济也没有范围不经济;
当 $SC < 0$，表示存在范围不经济

Summary

- Main concepts
  - economic cost, opportunity cost, sunk cost, total cost, fixed cost, variable cost, marginal cost, average cost
- Cost in the short run
  - diminishing marginal returns
  - average-marginal relationship
- Cost in the long-run
  - user cost of capital
  - isocost line
  - producing a given output at minimum cost
  - expansion path
  - long-run average-marginal relationship

Summary (Con't)

- Long-run versus short-run cost
  - economies of scale and diseconomies of scale
  - Increasing returns to scale versus economies of scale
  - measurement of economies of scale
  - relationship between short-run and long-run cost
  - product transformation curve
- Economies of scople
  - economies of scope and diseconomies of scope
  - degree of economies of scope

Exercises

1. Constantine purchased 100 shares of IBM stock several years ago for $150 per share. The price of these shares has fallen to $55 per share. Constantine's investment strategy is "buy low, sell high." Therefore, he will not sell his IBM stock until the price rises above $150 per share. If he sells at a price lower than $150 per share he will have "bought high and sold low." Constantine's decision:
   a. is correct and shows a solid command of the nature of ~~opportunity cost~~. 忽视了机会成本，而过于关注沉没成本
   b. is incorrect because the original price paid for the shares is a sunk cost and should have no bearing on whether the shares should be held or sold.
   c. is incorrect because when the price of a stock falls, ~~the law of demand~~ states that he should buy more shares.
   d. is incorrect because it treats the price of the shares as an ~~explicit cost~~.

Exercises (Con't)

2. Which of the following statements demonstrates an understanding of the importance of sunk costs for decision making?

这里提到的学费是已经支付且不可收回的沉没成本。
决策应该基于是否继续学习对未来收益的影响，而不是学费这个沉没成本。
因此，这个陈述没有正确理解沉没成本。

   I. "Even though I hate my MBA classes, I can't quit because I've spent so much money on tuition."
   II. "To break into the market for soap our firm needs to spend $10M on creating an image that is unique to our new product. When deciding whether to develop the new soap, we need to take this marketing cost into account."

这里提到的 $10M 是未来需要支出的成本，而不是已经发生的沉没成本。
未来决策中应该考虑这些成本的影响，这是正确的成本分析逻辑。
因此，这个陈述正确理解了成本分析，但不涉及沉没成本。

   a. I only
   b. II only
   c. Both I and II
   d. Neither I nor II

Exercises (Con't)

3. In a short-run production process, the marginal cost is rising and the average variable cost is falling as output is increased. Thus,
   a. average fixed cost is constant.
   b. marginal cost is above average variable cost.
   c. marginal cost is below average fixed cost.
   d. marginal cost is below average variable cost.

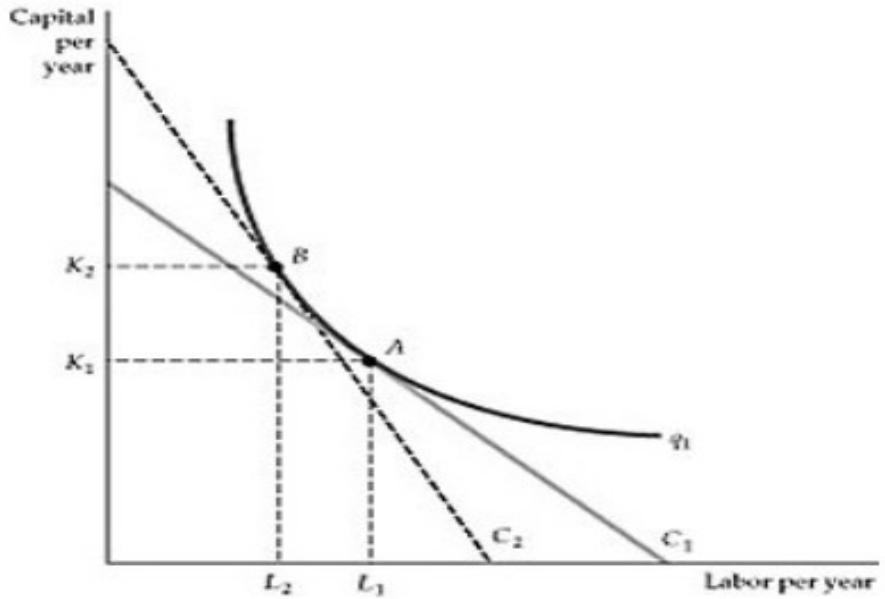
Exercises (Con't)

4. If a factory has a short-run capacity constraint (e.g., an auto plant can only produce 800 cars per day at maximum capacity), the marginal cost of production becomes _____ at the capacity constraint.
   a. infinite
   b. zero
   c. highly elastic
   d. less than the average variable cost

边际成本为生产额外单位的成本

当达到生产能力极限时，试图进一步增加产量会导致资源过度利用（如机器过载、劳动力超时工作），从而使边际成本变得极高

Exercises (Con't)



5. Refer to the above figure, which of the following changes cases the move from A from B?
   a. The price of one of the inputs
   b. the productivity of inputs
   c. the quantity to be produced
   d. the budget of the producer

投入品生产率的提高会影响等产量曲线的形状，以某一个点旋转
如果投入品的价格发生变化，会导致等成本线的斜率改变，而不是等成本线的平行移动
产量的变化会改变等产量曲线的位置，可能会需要新的生产组合

Exercises (Con't)

6. A firm employs 100 workers at a wage rate of \$10 per hour, and 50 units of capital at a rate of \$21 per hour. The marginal product of labor is 3, and the marginal product of capital is 5. The firm:
   a. is producing its current output level at the minimum cost.
   b. could reduce the cost of producing its current output level by employing more capital and less labor.
   c. could reduce the cost of producing its current output level by employing more labor and less capital.
   d. could increase its output at no extra cost by employing more capital and less labor.
   e. Both b and d are true.

Exercises (Con't)

7. Assume that a firm's production process is subject to increasing returns to scale over a broad range of outputs. Long-run average costs over this output will tend to:
   a. increase.
   b. decline.
   c. remain constant.
   d. fall to a minimum and then rise.

Exercises (Con't)

8. At every output level, a firm's short-run average cost (SAC) equals or exceeds its long-run average cost (LAC) because:
   a. diminishing returns apply in the short run.
   b. returns to scale only exist in the long run.
   c. opportunity costs are taken into account in the short run.
   d. there are at least as many possibilities for substitution between factors of production in the long run as in the short run.
   e. none of the above.

长期内，生产要素的替代性更强（如资本和劳动力的替代），使生产者能够更灵活地调整以达到最低成本。这解释了为什么 LAC 是 SAC 的包络线。

Exercises (Con't)

9. When a product transformation curve for a firm is bowed inward, there are _____ in production.
   a. economies of scope
   b. economies of scale
   c. diseconomies of scope
   d. diseconomies of scale